Lab:10

Automated Multi-Format Sales Report Pipeline and Delivery

INTRODUCTION:

This project implements an automated pipeline to generate sales reports in multiple formats (Excel, HTML, PDF) and optionally deliver them via email. The system processes sales data, performs analysis, and produces consistent reports for business insights.

THEORY:

Questions:

1. Explain the role of each step in the automated report pipeline: Extract → Transform → Generate → Deliver.
2. How does the pipeline ensure that Excel, HTML, and PDF reports are consistent in content?
3. Modify the pipeline to include a bar chart of monthly revenue in the PDF repor
4. Explain how you would secure the email credentials when deploying this pipeline in production.

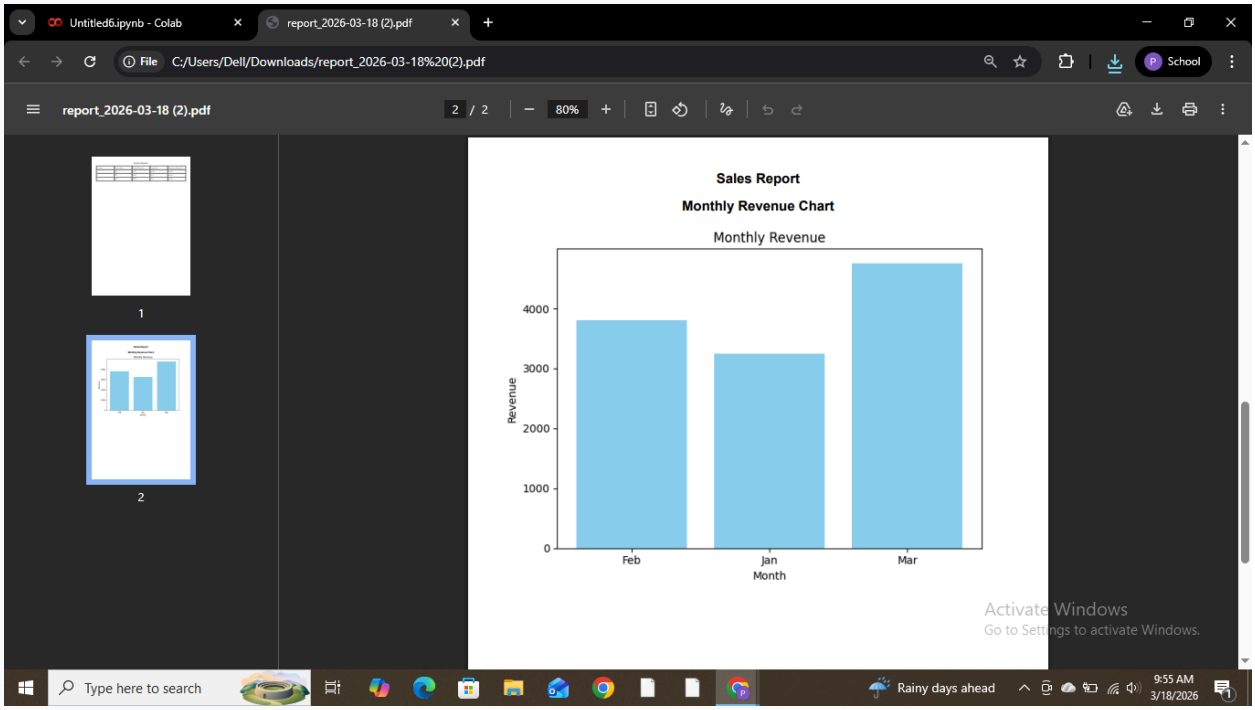
Answers:

1. Role of Each Step (Extract → Transform → Generate → Deliver)

- Extract: Collects raw sales data from source (file/database).
- Transform: Processes data by calculating revenue, summarizing totals, and cleaning.
- Generate: Creates reports in Excel, HTML, and PDF formats.
- Deliver: Sends the generated report to users via email.

2. Ensuring Consistency Across Reports

All reports (Excel, HTML, PDF) are generated from the same transformed dataset (summary DataFrame), so values like total revenue and units remain identical in every format.

3. 

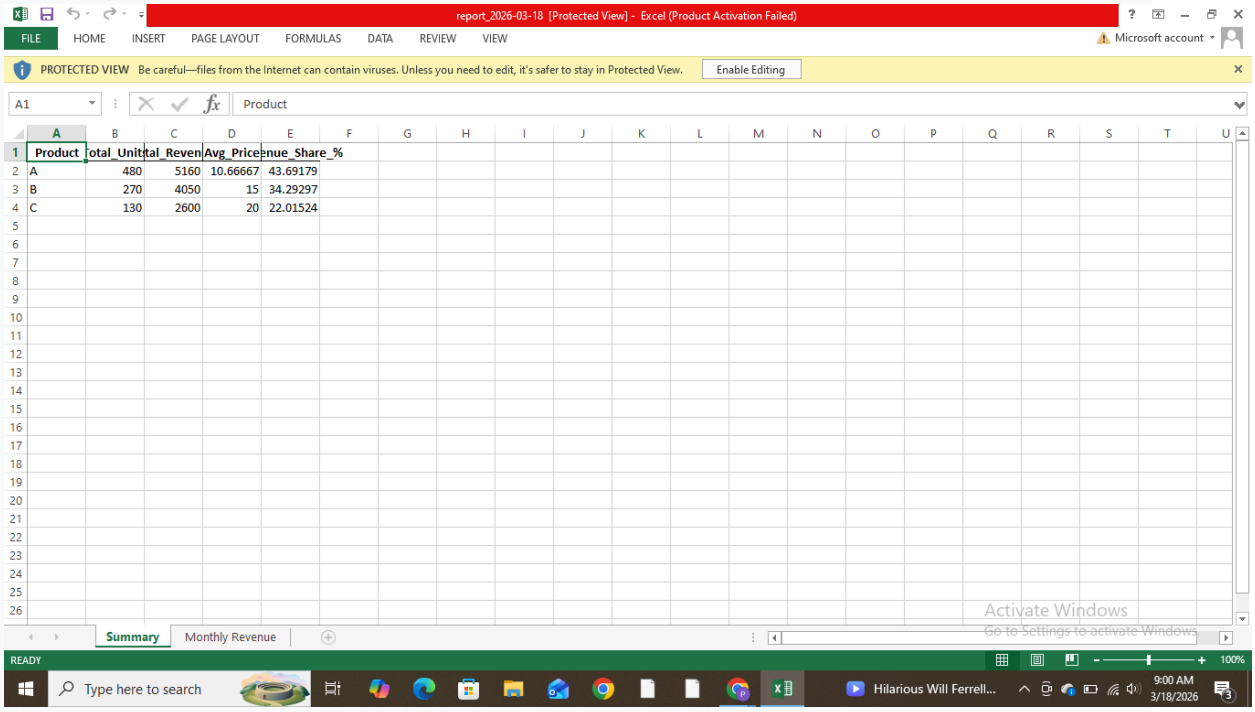
4. Securing Email Credentials

- Use environment variables instead of hardcoding passwords
- Use App Passwords (Gmail) instead of main password
- Store secrets using secure managers (e.g., AWS Secrets Manager)
- Avoid sharing credentials in code repositories

1. Generated Reports Folder

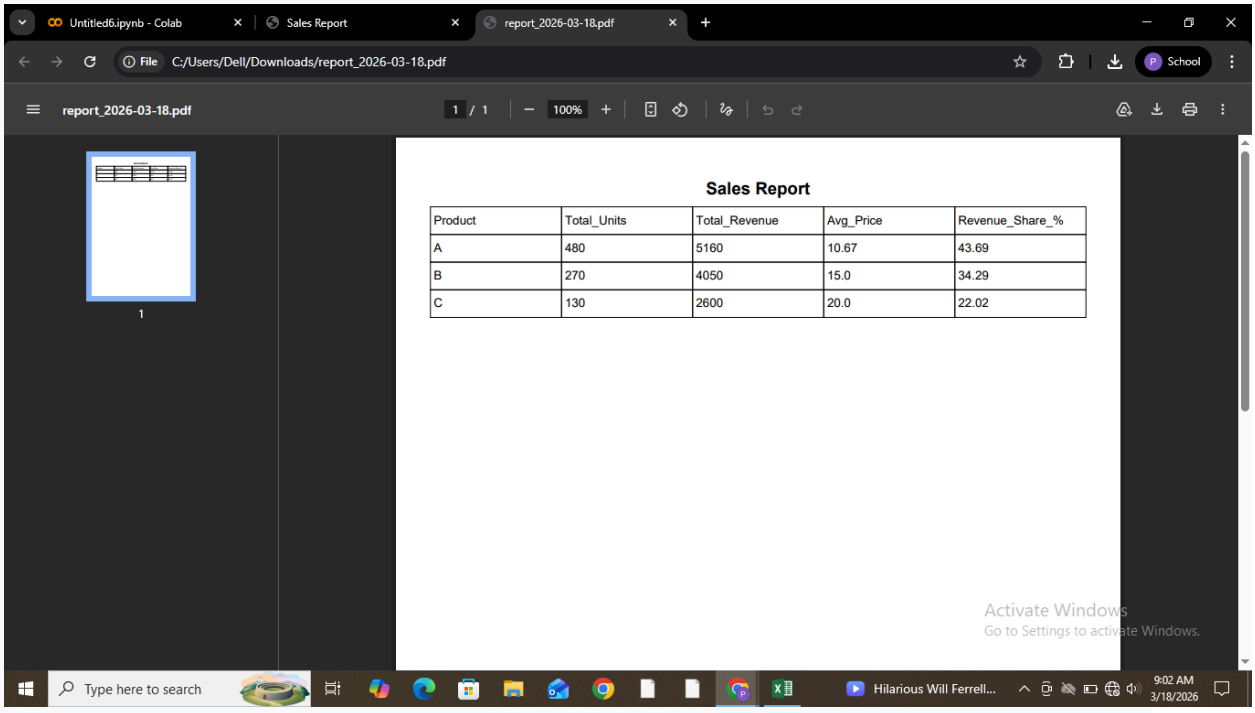
The system successfully generated all required report formats.
The screenshot below shows the `reports` folder containing:

- Excel file :



Product	Total_Units	Total_Revenue	Avg_Price	Revenue_Share_%
A	480	5160	10.66667	43.69179
B	270	4050	15	34.29297
C	130	2600	20	22.01524

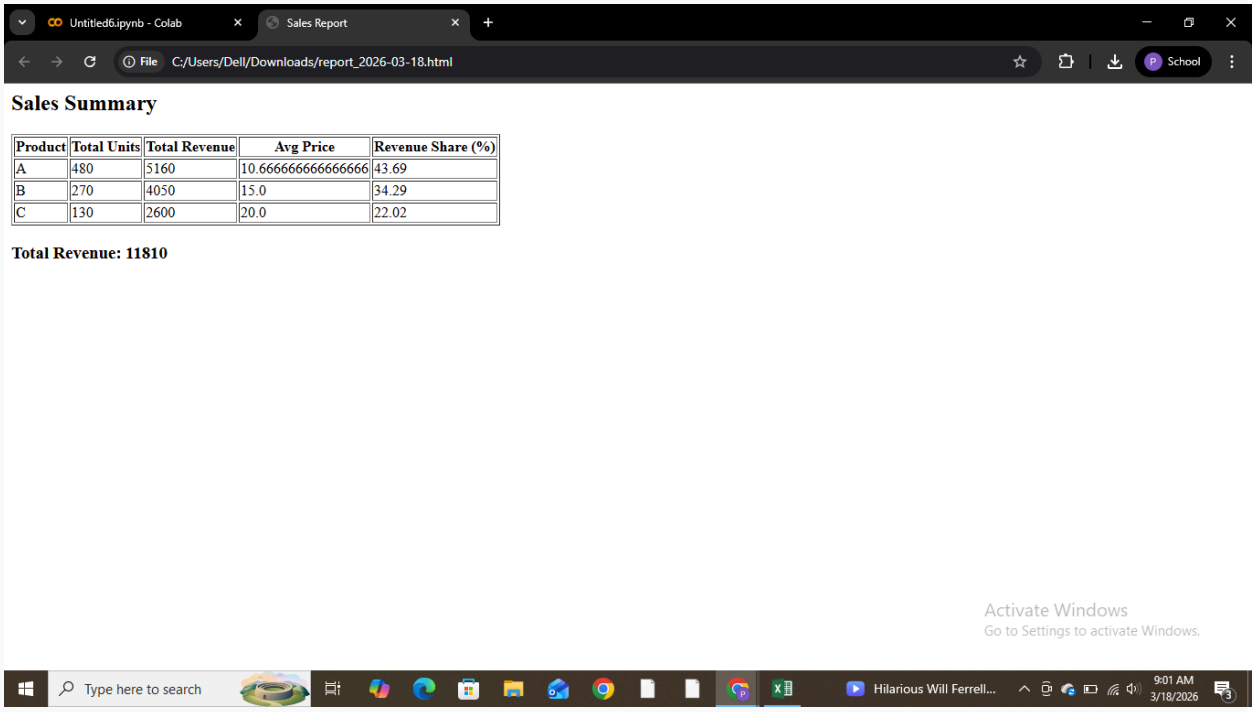
- pdf:



Sales Report

Product	Total_Units	Total_Revenue	Avg_Price	Revenue_Share_%
A	480	5160	10.67	43.69
B	270	4050	15.0	34.29
C	130	2600	20.0	22.02

-html file:



CONCLUSION:

The pipeline successfully automates report generation and delivery. It ensures consistency across formats and reduces manual effort. This approach can be extended for real-world business reporting systems.